

Analysis of Historical Environmental Data Collected by Ocean Gliders in the Gulf of Saint Lawrence



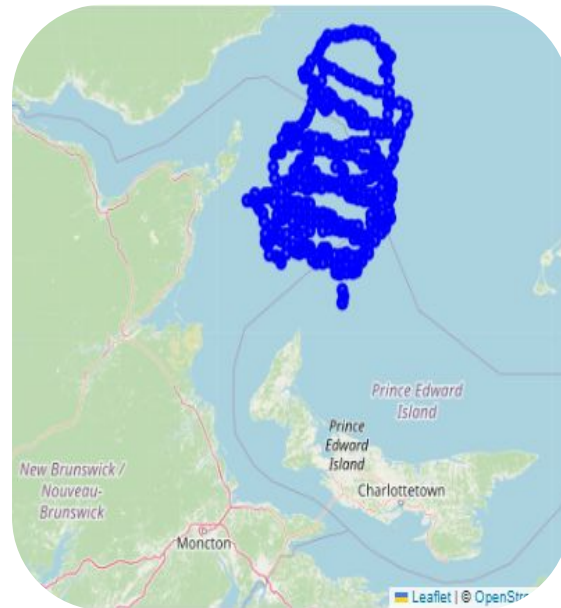
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Understanding oxygen depletion in the Gulf of Saint Lawrence

- **Context:** The Gulf of Saint Lawrence has witnessed declines in oxygen concentration levels
- **Concern:** risk of hypoxia (oxygen concentration less than $60 \mu\text{mol/kg}$) and low-oxygen events which threaten marine life and alter ecosystems.

This project analyzes historical underwater glider data between 2018 and 2023. The goal is to extract insights that support conservation, policy decisions, and sustainable marine practices.



“Oceanic oxygen is essential for marine ecosystems & biodiversity.”

Research Question

How do oxygen concentration levels in the Gulf of Saint Lawrence vary over time and space, and what are the factors that influence these changes?



Temporal Trends

- Oxygen concentration over years or seasons?
- Weekly, Hourly or Daily variations?
- General trend of Oxygen Concentration?

Can we detect low oxygen timelines?



Spatial Variations

- Any Regional Oxygen Concentration patterns over time ?
- Any changes in Oxygen Concentration levels across different regions?

Are certain regions prone to oxygen depletion?



Other Factors (Environmental Drivers)

- Relation between oxygen concentration and individual or joint environment factors?
- Any thresholds that trigger oxygen decline?

Do certain variables reduce oxygen concentration?

Exploratory Data Analysis

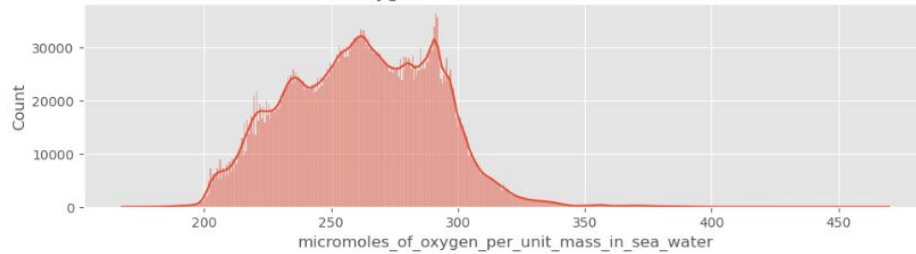


Dataset Description

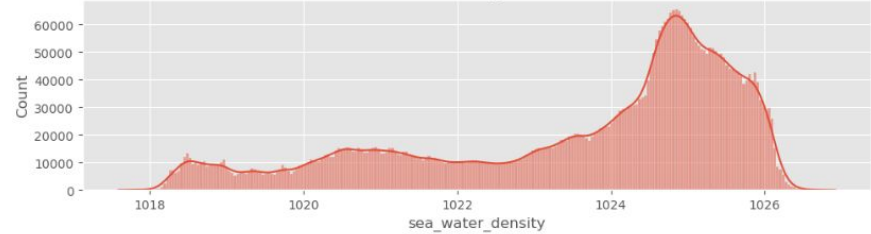
- Glider DataSet (9 of 15)
- Excluded Ship (Bottle) DataSet
- 4,530,334 rows, 323 days total
- 2018-2023, June to December
- Unbalanced proportions:
 - 10% is 2021, <3% nov, sep, .. etc
- Few months recorded (6-9,11-12)
- No missing/null values
- No duplicates
- Many Outliers: depth, density, salinity and oxygen concentration
- Regional Data Imbalance
 - Shad: 11%, Scotia: 33%
Cabot: 47%, Fundy: 1% Sambro: 8%

Data Distribution

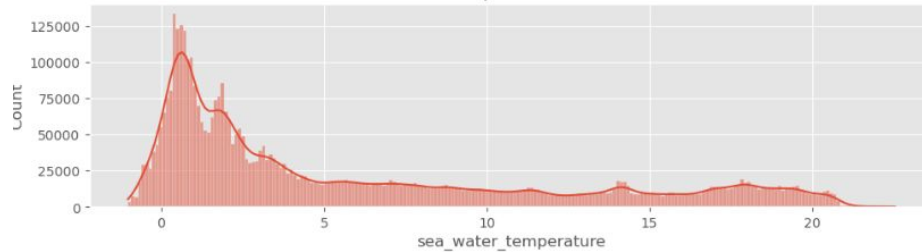
Micromoles Of Oxygen Per Unit Mass In Sea Water Distribution



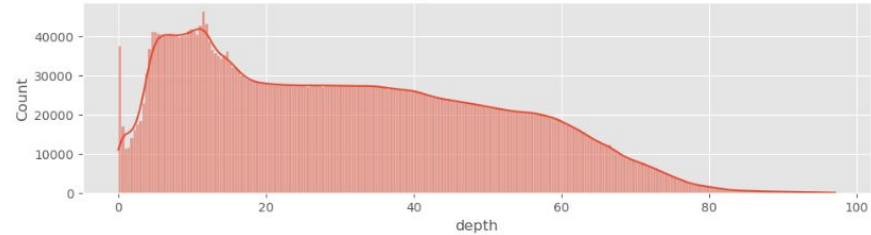
Sea Water Density Distribution



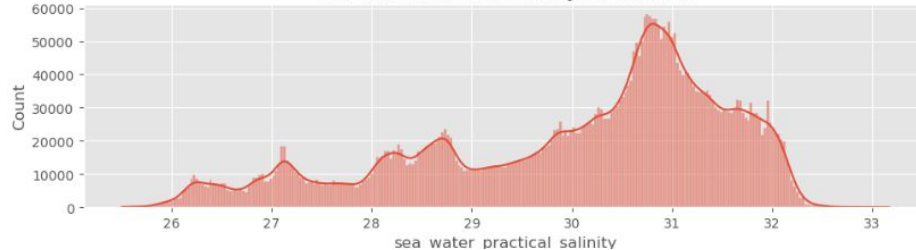
Sea Water Temperature Distribution



Depth Distribution



Sea Water Practical Salinity Distribution



Data Distribution:

Oxygen: 167.5 to 470 ($\mu\text{mol/kg}$)

Salinity: left-skewed, 25.47 to 33.17 (PSU)

Density: left-skewed, 1017.5 to 1026.9 (kg/m^3)

Temperature: right-skewed, -1 to 22.5 ($^{\circ}\text{C}$)

Depth: 0 to 97.12 (meters)

Data Cleaning

Data Cleaning approach:

- Corrected localized errors using neighbouring values, and recurrent patterns
- Removed illogical negative values (i.e. negative depth or negative oxygen concentration)
- Used IQR to remove lower & upper bound outliers

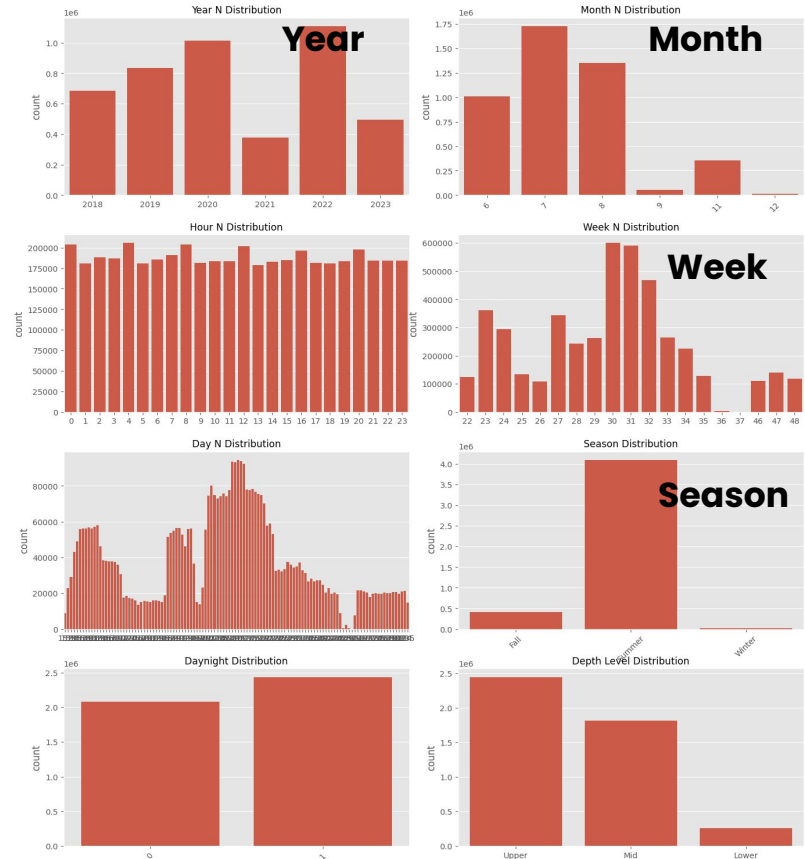
Result:

- 4,515,767 samples
- No hypoxic samples

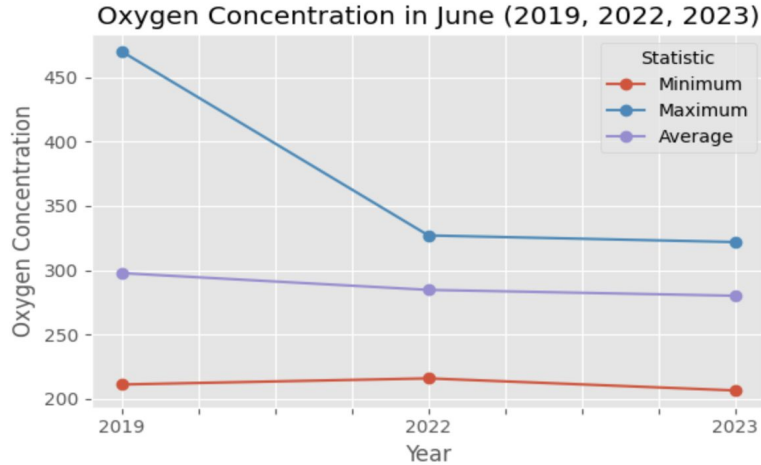


Temporal Patterns: EDA Insights

- Cannot compare overall **yearly** average (data proportions vary significantly)
- Similarly the vast majority is **summer**-data; on average it has lower oxygen while **winter** has the highest.
- More specifically, the **monthly** average decreases from June to September, then increases again.
- **Weekly** mean oxygen concentration further proves that there is decline from week 22 to 34 and rapid increase from week 35 to 45.

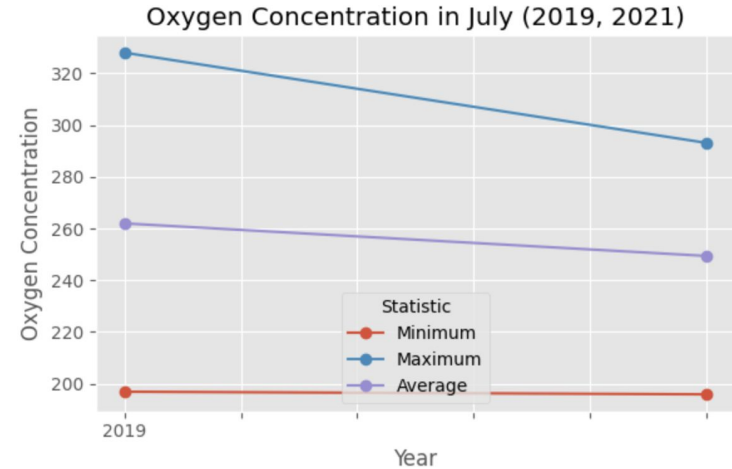


Temporal Patterns: EDA Insights



Oxygen concentration in June 2019, 2022, 2023:

- Sharp decrease in max. oxygen between 2019 & 2022 (over $100 \mu\text{mol/kg}$ less)
- Slight decrease in oxygen 2022 to 2023 in terms of minimum, average and maximum.



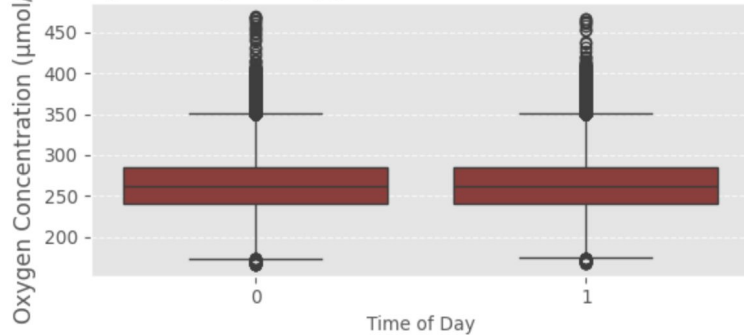
Oxygen concentration in July 2019, 2021:

- Decrease in maximum oxygen recorded between 2019 & 2021 ($\sim 40 \mu\text{mol/kg}$ less)
- Very slight decrease in terms of minimum, oxygen concentration recorded ($< 5 \mu\text{mol/kg}$)

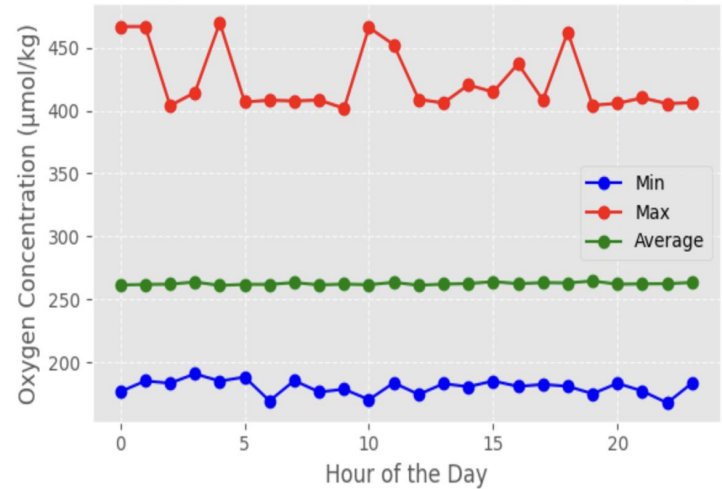
Temporal Patterns

	Day/Night	Min	Max	Average
0	Night	167.520615	469.951770	262.322288
1	Day	169.143424	466.173638	262.400119

Day vs. Night Oxygen Concentration Distribution



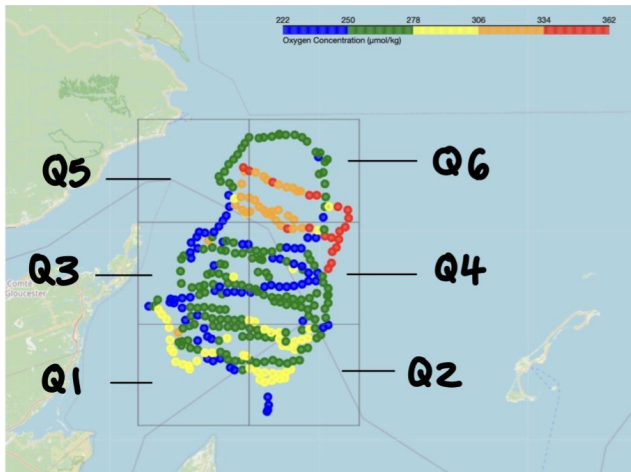
Oxygen Concentration Throughout the Day



No difference in Oxygen levels during the day (**Day vs. Night**)

Maximum oxygen levels peak (increase by 50 $\mu\text{mol/kg}$) at certain hours; before sunrise/noon and at sunset

Spatial Patterns: EDA Insights



- Average oxygen concentration remains relatively constant over time, over different quadrants/locations

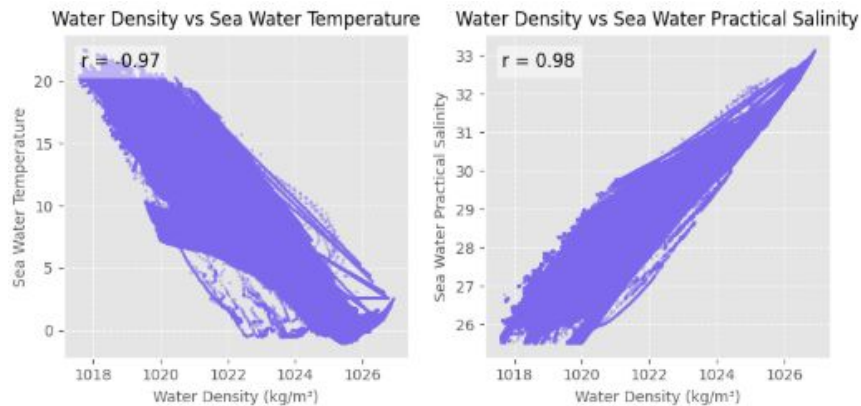
Note: Q5/Q6 unreliable – not enough data

Location Trends:

- No direct relation between oxygen concentration and longitude/latitude
- Varying average oxygen concentrations – not in distinct clusters

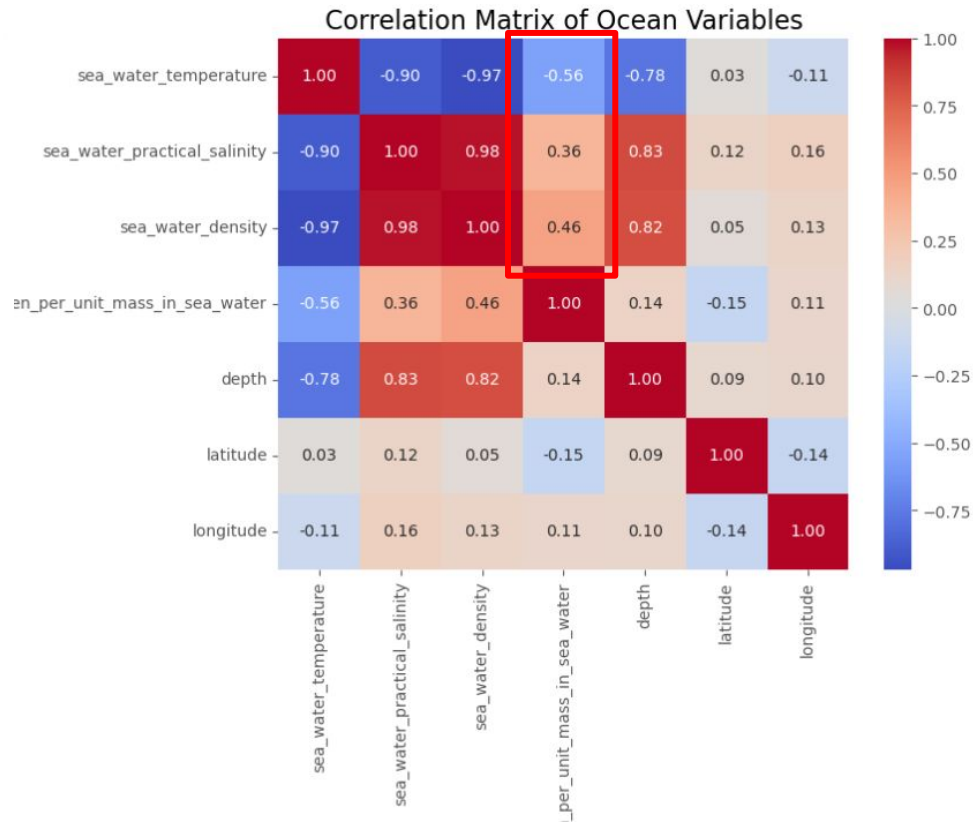


Correlations and Linear Patterns: EDA Insights

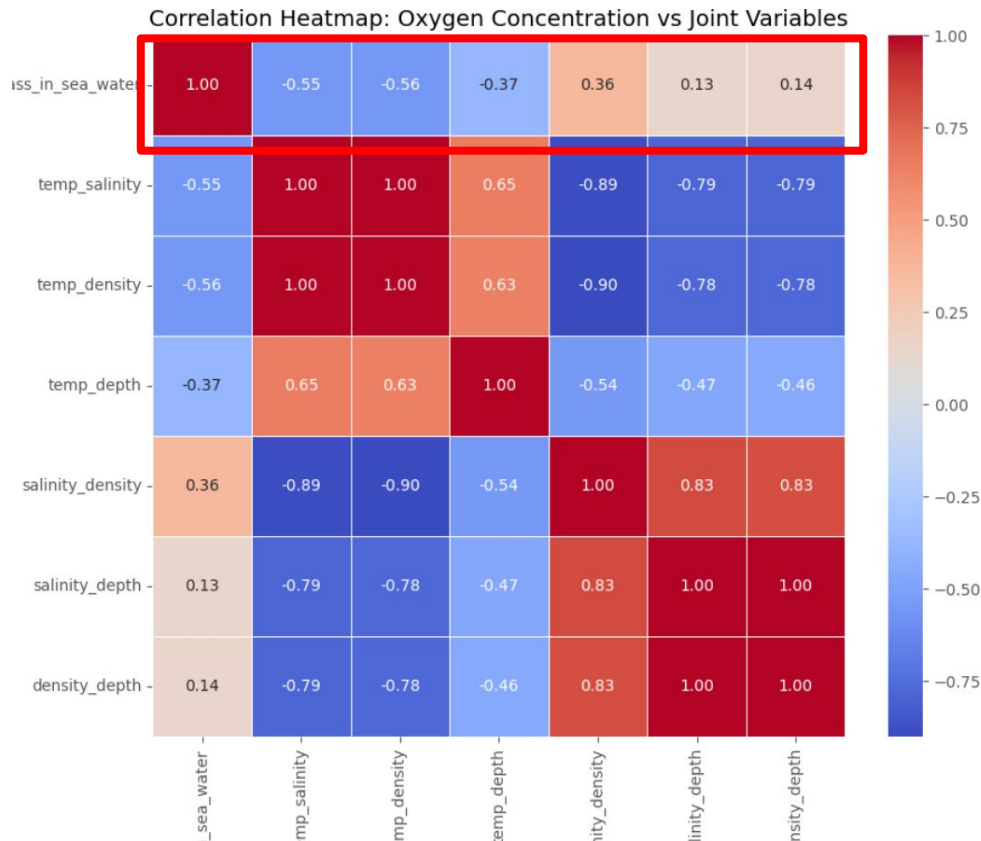


Linear relationship / Correlation between:

- Temperature and Density (-0.97)
- Salinity and Density (+0.98)
- Temperature and Salinity (-0.90)
- Depth & Salinity/Density (0.83/0.82)



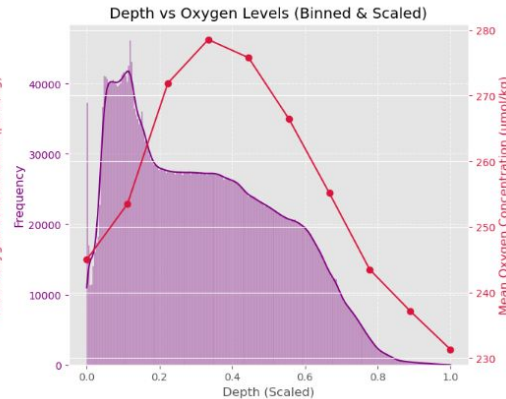
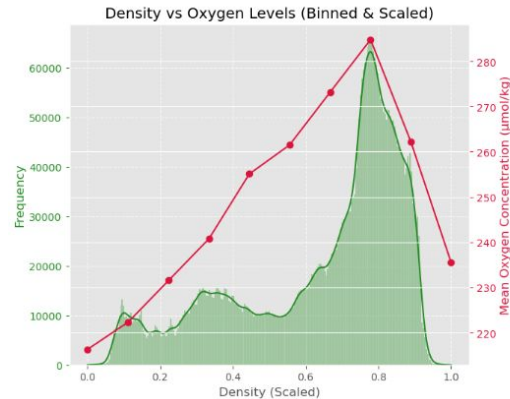
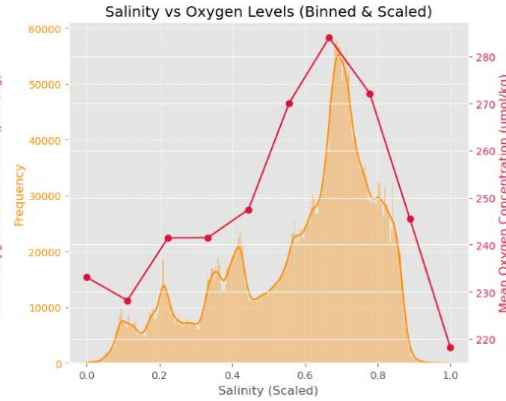
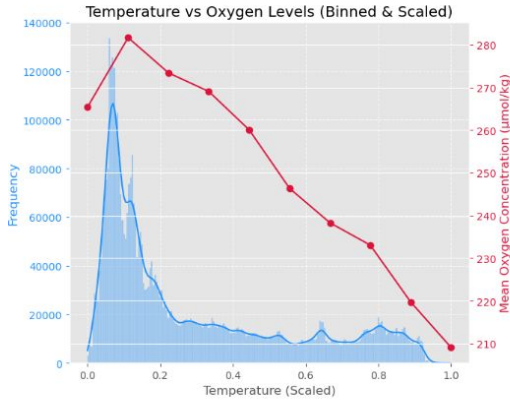
Interactions: EDA Insights



Confounding variables & Interactions:

- Strong correlation
 - temp_density & temp_salinity
 - Salinity_depth & density_depth
- Oxygen concentration
 - Highest: temp_density

Environmental Factors



Oxygen concentration peaks when:

- Temperature is low ~1.33 to 3.69°C
- Density is high ~ 1024.12 to 1025.05 kg/m³
- Salinity is high ~ 30.10 to 30.87 PSU
- Depth is average ~ 29.14 to 38.85 m

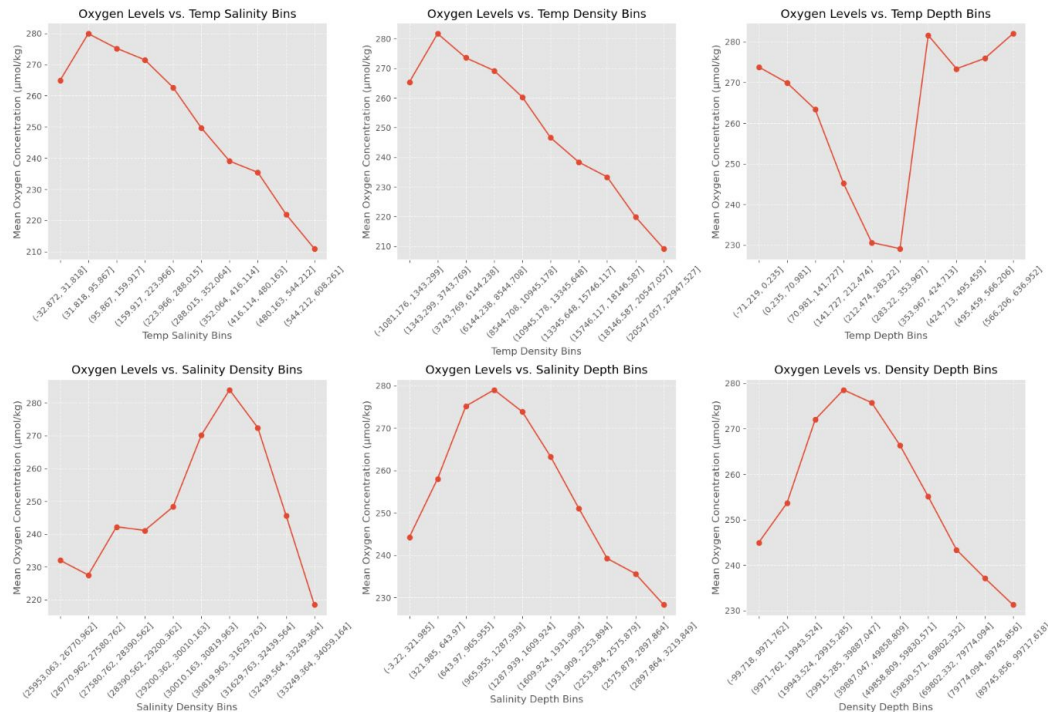
Henry's Law: oxygen solubility ↑

- ↓ temperature
- ↑ denser
- ↑ salinity conditions

Depth: oxygen concentration ↑

- 0-20m : photosynthesis
- 30-40m: aeration from winds/tides

Confounding Environmental Factors



Joint Variables

- Oxygen levels gradually increase until they peak for certain values of the joint variables. Then, they decrease again.
- In our case, the peak was at $\sim 280 \mu\text{mol/kg}$.
- An inverse pattern can be observed for Temp_depth where oxygen level dips as it approaches a certain value.



Statistical Models

Random Forest Regression & Linear Regression

Random Forest Regression



Data Processing

Categorical Variables	Created dummy variables
Numeric Variables	Log transformation for the target variable
	Square-root transformation to normalize variables
Feature Selection	Exclude density due to high correlation
Evaluation Approaches	GridSearchCV with 5-fold Cross-validation: to find hyperparameters
Data Splitting	Training 70% Test 30%

Model Construction

Features	Metrics & Results
Temperature Salinity Month_n Depth hour	Residual analysis - Histogram: normal (peak at 0) - scatter plot: random scatter - QQplot: aligned with straight line R-square: 0.9317/0.6173 AIC: -2253.81/-784.52 RMSE: 0.0246/0.0640

Best Model: Linear Regression



Data Processing

Categorical Variables	Created dummy variables
Numeric Variables	Log transformation np.log on the model to find the better R-square
Feature Selection	Trial and Error - Different Models/Different Features - Best R^2 / lowest AIC, skewness, kurtosis scores
Evaluation	Cross-validation
Data Splitting	Training 80%, Test 20%

Model Construction

Features	Metrics & Results
Month_n Depth Temperature Salinity	Residual Analysis - Histogram - QQ plot - Comparison from the other model, found the best fit. R - squared: 0.748 AIC value: -5.26×10^6 RMSE: 0.04914

Results



```
=====
                        OLS Regression Results
=====
Dep. Variable:      np.log(micromoles_of_oxygen_per_unit_mass_in_sea_water)  R-squared:                0.748
Model:              OLS                                                    Adj. R-squared:           0.748
Method:             Least Squares                                           F-statistic:              7.020e+05
Date:               Tue, 01 Apr 2025                                         Prob (F-statistic):       0.00
Time:               02:46:29                                                 Log-Likelihood:           2.6340e+06
No. Observations:   1652290                                                 AIC:                     -5.268e+06
Df Residuals:       1652282                                                 BIC:                     -5.268e+06
Df Model:           7
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	5.5965	0.000	2.52e+04	0.000	5.596	5.597
C(Month_n)[T.7]	-0.0603	0.000	-450.507	0.000	-0.061	-0.060
C(Month_n)[T.8]	-0.0963	0.000	-622.220	0.000	-0.097	-0.096
C(Month_n)[T.9]	-0.1571	0.000	-485.206	0.000	-0.158	-0.156
C(Month_n)[T.11]	5.859e-16	4.52e-19	1296.177	0.000	5.85e-16	5.87e-16
C(Month_n)[T.12]	-5.524e-16	2.17e-19	-2542.116	0.000	-5.53e-16	-5.52e-16
depth	-0.0334	0.000	-197.222	0.000	-0.034	-0.033
np.log(sea_water_temperature)	-0.0066	6.39e-05	-103.115	0.000	-0.007	-0.006
sea_water_practical_salinity	-0.0047	0.000	-554.715	0.000	-0.005	-0.004
sea_water_density	0.1452	0.000	691.120	0.000	0.145	0.146

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=====
```

- $p_values < 0.05 \Leftrightarrow$ all variables are statistically significant
- $R^2 = 0.748 \Leftrightarrow$ The model can explain 74.8 % of the variability of the data
- Skewness: 1.431
- Kurtosis: 9.312
- $AIC = -5.26 \cdot 10^6$ (Avoids overfitting)

Coefficients:

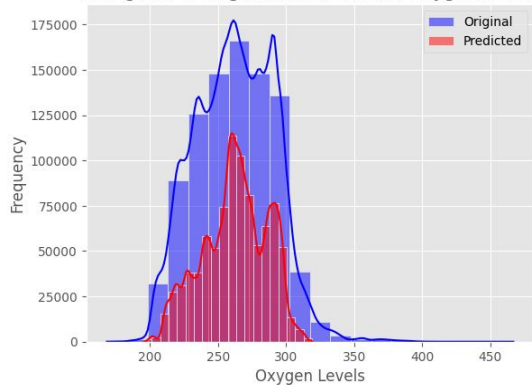
Depth, temp, salinity, month: **negative**

Density: **positive**

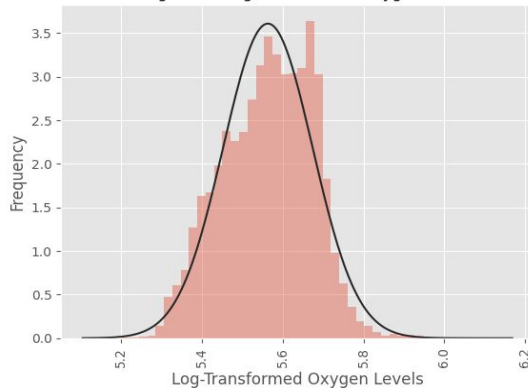
Results



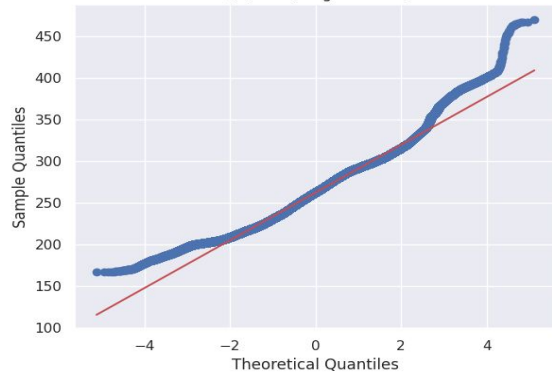
Histogram of Original and Predicted Oxygen Levels



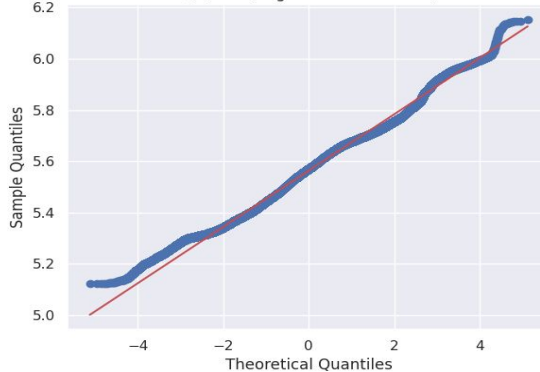
Histogram of Log-transformed Oxygen Levels



Q-Q Plot (Original Data)



Q-Q Plot (Log-Transformed Data)



**Applied Log Transformation
to the target variables
(Oxygen Concentration)**

QQ-plot/histogram appears more gaussian after log transformation

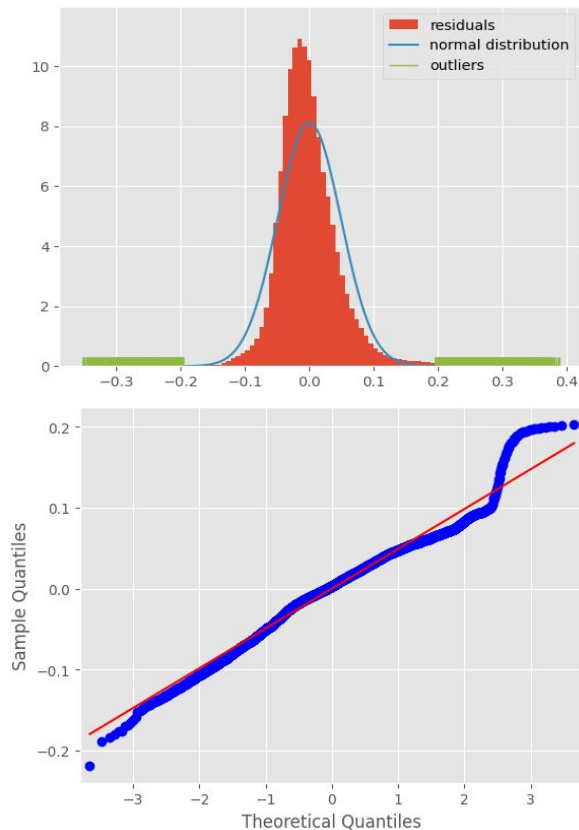
Results



Residual and QQ Plot of the Model

```
=====
Omnibus:                    537834.054    Durbin-Watson:                0.073
Prob(Omnibus):              0.000        Jarque-Bera (JB):            3306763.831
Skew:                      1.431         Prob(JB):                  0.00
Kurtosis:                  9.312         Cond. No.                  4.36e+17
=====
```

- **Skewness and Kurtosis**
 - Minimal tail heaviness
 - Mostly symmetric model
- **QQ Plot**
 - Mostly follows normal distribution
 - Right skewness



Interpretation



Log (Oxygen Concentration):

Intercept: 5.5965

Month_n

- July: - 0.0603
- August: - 0.0963
- September: - 0.1571
- November: 5.859×10^{-16}
- December: -5.524×10^{-16}

Depth: - 0.0

Log(temp.): - 0.00

Salinity: - 0.08

Density: 0.1452

Linear Regression Interpretation:

- Base Oxygen Level (*Intercept* 5.5965)
 - The baseline oxygen concentration is 5.5965 when other variables are at reference levels.
- Monthly effect:
 - Summer/ Fall Months: significant reduction Sept: strongest changes
 - Nov/Dec: negligible effect
- **↑ depth (1m): ↓ 3.4% oxygen concentration**
- **↑ temperature (1%) : ↓ 0.0066% oxygen concentration**
- **↑ salinity(1PSU): ↓ 8.47% oxygen concentration**
- **↑ density (1 kg/m³): ↑ 14.52% oxygen concentration**



Next Steps

*Continuing oxygen preservation through directed
research studies and interventions*



Recap

1. Oxygen levels peaks are higher around before sunrise, noon, sunset and midnight.

- Collecting information about **tides** (timing, strength) may help identify the degree to which they introduce oxygen-rich water.
- As **wind** periodically shifts at certain times of the day (causing warm air to rise from the ocean, and enhancing oxygen diffusion from the atmosphere into the water), we suggest collecting information about wind speed and direction, as well as the surface's **weather**.
- **Nutrient runoff** fuels algal blooms, which temporarily boost **daytime oxygen** but cause nighttime hypoxia as dead algae decompose (NOAA, USA Gov.)

2. Deeper waters have less oxygen

- Surface waters have optimal light availability (midday peak) and nutrient supply which causes them to be oxygenated by **photosynthesis** and **atmospheric exchange**.
- Phytoplankton blooms produce oxygen at the surface, while sinking organic matter consumes it at depth.

3. Salinity decreases oxygen levels:

- Cooler, fresher water contains more oxygen than warm, salty water



Future Research Suggestions:

- Collect data on tides, wind, weather, and oceanic photosynthesis as well as maritime traffic to identify other oxygenation factors.
- Study plankton species to identify which ones to cultivate or reintroduce for balanced oxygen production.
- Continue research to help prevent and manage hypoxic (low-oxygen) events.



Conservation Recommendations:

- Reduce nutrient pollution via better wastewater treatment and ecosystem protection.
- Limit fertilizer use to prevent oxygen depletion.
- Expand reefs and seabeds for natural filtration and oxygen generation.
- Support ocean-friendly energy, low-emission shipping, and traffic regulation.
- Use artificial aeration in critical zones to mimic natural oxygenation

The background is an underwater scene. The top half shows the water surface with ripples and light reflecting off it. Below the surface, the water is a deep blue, and several bright rays of sunlight penetrate from above, creating a serene and slightly ethereal atmosphere. Small particles or bubbles are visible in the water.

Thank You